

The Use of Regression Models to Predict NBA Teams' Win Distributions

Grzegorz Ruchała, Bartłomiej Węgrzyn

The aim of this paper is to present the process of building various regression models using the R programming language. The study will employ linear regression, multiple regression, and Poisson regression models. Based on statistical data from previous NBA seasons made available by the league, these models will be used to estimate the number of wins achieved by individual teams in subsequent seasons.

The leaps and MASS packages will be applied, along with tools that enable the automation of variable selection. In addition, Poisson regression models will be developed using the aforementioned packages. The paper will also demonstrate how statistical methods can be used to determine which of the developed models achieves the highest predictive accuracy.

Data Preparation

We will begin by preparing the data and dividing it into training and test sets. The training set will consist of combined data from the 2015/16, 2016/17, and 2017/18 seasons, while the test set will contain data from the 2018/19 season. We remove columns that will not be relevant when building the models, either because of their format, such as team names and arena names, or because they contain redundant information. For example, we remove the number of games lost because we use the number of games won.

We also remove outliers lying outside the 0.02 and 0.98 quantiles. These consist of two lower observations and one upper observation that could significantly distort the results due to the record-breaking 73-9 win-loss record achieved by the Golden State Warriors in the 2015/16 season, one of the best records in basketball history.

Model Development

First, we will create a linear regression model. Linear regression involves fitting a linear relationship between the dependent and independent variables, expressed as the linear function $y = \beta_0 + \beta_1 x$.

In this model, we select MOV, or margin of victory, as the independent variable. It represents the average point differential per game. The higher the value, the less evenly matched the game and the greater the final score difference.

In the multiple regression model, we use all 21 variables remaining after the initial data processing. Examples include the average player age, possessions per game, known as pace, free throws and three-point attempts as a percentage of all field-goal attempts, points scored per 100 possessions (ORTg), and points allowed per 100 possessions (DRtg).

Model Development Using R Packages

We will now build multiple regression models using R packages. First, we use the MASS package, which automatically adds or removes variables to minimize the Akaike Information Criterion (AIC). The stepAIC function simplifies the model without significantly reducing its accuracy. The final model contains only three variables: MOV, average player age, and SRS, a statistic representing point differential adjusted for opponent strength.

The second model uses the leaps package, which searches for the best set of variables based on a specified metric—in our case, adjusted R^2 . The final model includes six variables: average player age, projected wins based on points scored and allowed, SOS, which reflects opponent difficulty and is especially relevant because NBA teams do not play every opponent equally often, three-point attempt percentage, effective field-goal percentage, which accounts for different shot values, and, interestingly, average attendance per game.

Table 1 - Numerical Performance Metrics for the Models

	R^2	Adjusted R^2	AIC	RMSE	MAPE
Linear Regression	0.917	0.916	437.65	3.10	6.87%
Multiple Regression	0.931	0.907	462.11	3.25	6.43%
MASS Model	0.923	0.920	435.22	3.02	6.76%
leaps Model	0.926	0.921	437.28	3.01	6.42%

Poisson Regression

The Poisson distribution describes the probability of events occurring within a given time interval when the expected number of events is known and the events themselves are independent.

Under certain assumptions, the wins of individual teams can be treated as a sequence of independent events whose frequency depends on specific factors. The final outcome can only be a non-negative integer that can be interpreted directly:

0 - in the case of a loss,
1 - in the case of a win.

Therefore, we can see that the response variable follows a Poisson distribution:

$$P(Y = y_i | X_i) = \frac{\exp(-\lambda) \lambda^{y_i}}{y_i!}$$
$$\ln(\lambda_i) = X_i \beta$$
$$\mathbb{E}[X] = \lambda$$

We will now create two models based on Poisson regression.

The first, similarly to the multiple regression model using the MASS package, will be based on the variables Age, MOV, and SRS.

The second will be created using the stepAIC function and will depend only on the MOV variable.

Comparison of All Models

To evaluate the accuracy of each model, we will compare the predicted number of wins with the actual number of wins in the following season. For this purpose, we will use the results from the season immediately after the final season included in the models.

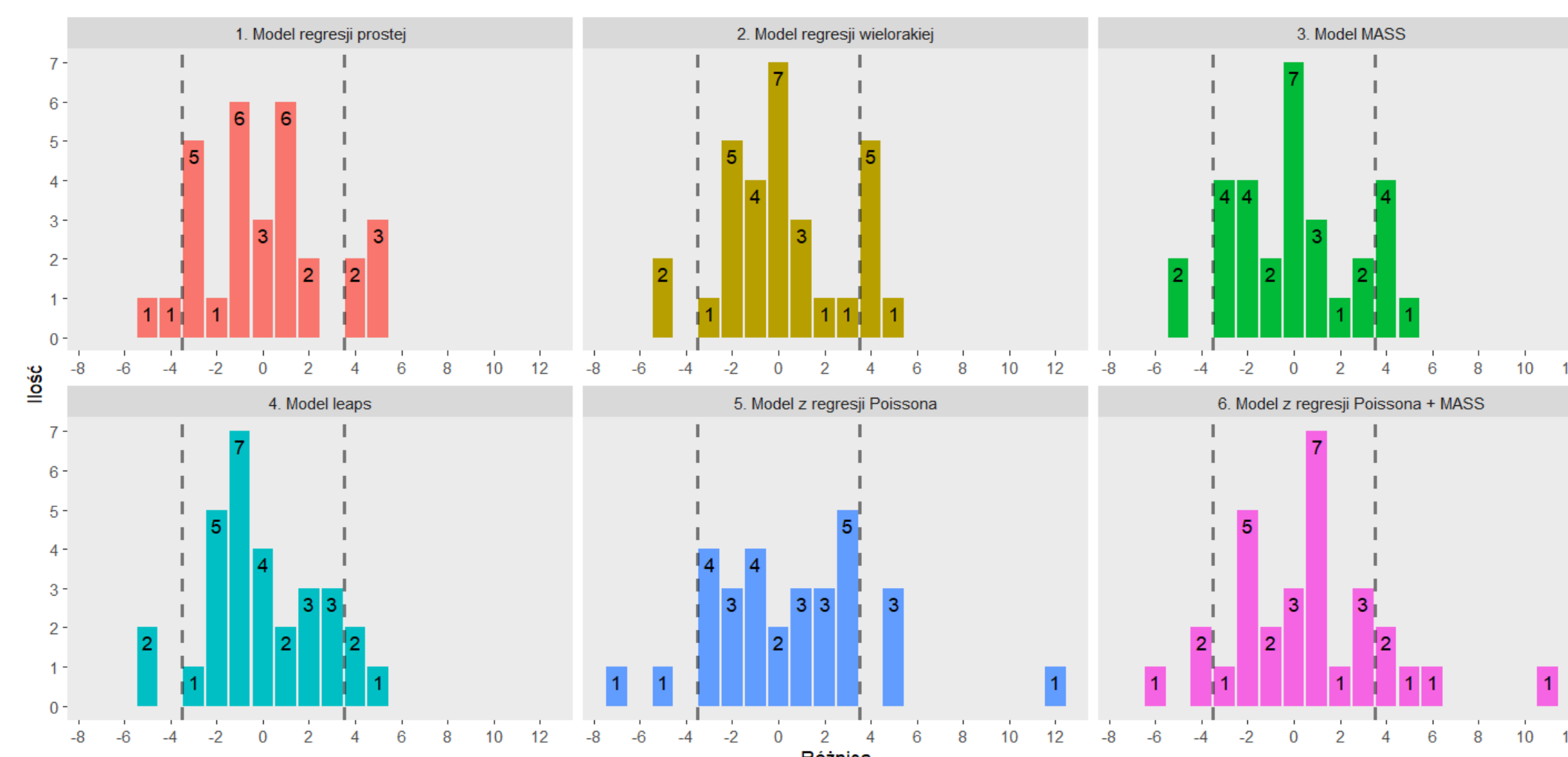


Figure 1 - Histograms of Prediction Errors

The histograms above show the distribution of prediction differences compared with the actual results.

We consider predictions satisfactory when the difference between the predicted and actual values is no greater than 3, for both underestimation and overestimation.

We can now characterize all of the models:

1. Linear regression - despite its simplicity, it predicts the values correctly,
2. Multiple regression - accurate predictions for a large number of cases; some predictions are significantly overestimated,
3. MASS model - an accurate and balanced model, with only a few predictions outside the satisfactory range,
4. leaps model - the only model showing a tendency to underestimate; only a few predictions fall outside the range,
5. Poisson regression - balanced, although it produces relatively few perfectly accurate predictions; one prediction differs greatly from the actual value,
6. Poisson regression + MASS - balanced and highly accurate; it also contains one outlying prediction.

Conclusions

Modeling sporting events is difficult because of the many different factors involved, most of which are related to human behavior. However, we can simplify the problem and use relatively simple mathematical methods to obtain predictions that are reasonably accurate for such a multidimensional problem. We can also observe a tendency for our models to overestimate the results.

Limitations and Challenges

One of the greatest limitations is the omission of the human factor represented by the draft. After each regular season, a lottery takes place, giving lower-ranked teams the right to select new players before the remaining teams. We are also unable to clearly identify and classify the most accurate of the six models developed. However, each of them is capable of modeling the problem with a certain degree of accuracy.

References

<https://www.nba.com/stats/teams>
<https://stats.oarc.ucla.edu/r/dae/poisson-regression/>
https://www.basketball-reference.com/leagues/NBA_2019.html
<https://www.statology.org/stepaic-r/>
<https://www.dataquest.io/blog/tutorial-poisson-regression-in-r/>

Koło Naukowe Matematyków, Katedra Matematyki Stosowanej,
ul. Warszawska 24, 31-155 Kraków
kontakt.sknmpk@gmail.com, dr Marcin Skrzyński
Opieka naukowa: Adrian Kowalski

